

# Surrogate Substitution Preserves PHI Detectability: A Multi-Detector Equivalence Study

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## Abstract

Structure-preserving de-identification replaces protected health information (PHI) with realistic same-type *surrogates*—“Anna S.” becomes “Maria S.”, not [NAME]—so that clinical text stays fluent and downstream tools keep working. But this only helps if the substitution does not itself corrupt the signal those tools rely on. We ask a narrow, testable question: *on the spans a de-identifier actually masks, can downstream PHI detectors still find the surrogate?* We introduce a paired, multi-detector evaluation protocol that (i) scores utility *only on masked spans*, decoupling *coverage* from *utility*; (ii) uses *equivalence testing* (TOST) rather than null-hypothesis significance testing, which is uninformative at our sample size (57k paired spans); and (iii) builds a surrogate-failure typology separating fixable generator defects from intrinsic detector limits. Across 11 detectors, 7 benchmarks, and 7 languages (1,750 documents), recall on masked spans moves from 76.1% to 74.9%—a change our equivalence test shows is *statistically equivalent to zero within a  $\pm 2$ -point margin* ( $p \approx 3 \times 10^{-9}$ ), with detector ranking preserved. The residual loss does not reflect detectors getting worse at PHI: it concentrates in *malformed and out-of-distribution* surrogates (truncation Chicago  $\rightarrow$  Illino, salience loss Cedars-Sinai  $\rightarrow$  Vidant). A redaction floor and an open-source surrogate baseline indicate the effect is a property of well-formed substitution, not of one tool. We release the evaluation subsets, scoring code, and an interactive dashboard at <https://custodianai.pages.dev> so the protocol can audit any structure-preserving transform.<sup>1</sup>

## 1 Introduction

De-identification of clinical text takes two forms—and only one keeps the text usable downstream. **Redaction** deletes or tags PHI (John Smith  $\rightarrow$  [NAME]), which is safe but destroys the fluency, layout, and distributional properties that downstream models and human readers depend on. **Structure-preserving** de-identification instead substitutes each PHI value with a plausible, same-type *surrogate* (John Smith  $\rightarrow$  Maria Lopez), keeping the document readable and machine-parseable [4]. The second family is increasingly attractive: it lets de-identified data flow into analytics, model training, and even second-pass detection without breaking pipelines built for real text.

The promise of structure preservation rests on an unstated assumption: *the surrogate carries the same downstream signal as the value it replaced*. If substitu-

tion silently degrades the very features a PHI detector, an NER model, or a clinical parser relies on, then “structure-preserving” is a misnomer—the transform would be quietly laundering PHI into forms tools can no longer see or handle, which is both a utility problem and, for a second-pass safety net, a privacy problem.

This assumption is rarely tested directly, and testing it well is harder than it looks. Three pitfalls recur:

1. **Coverage confounds utility.** A de-identifier that masks 60% of PHI and one that masks 95% cannot be compared on whole-document F1—the score conflates *how much* it masks with *whether what it masks stays usable*. The two must be measured separately.
2. **The large- $N$  significance trap.** With tens of thousands of paired spans, any non-zero difference is “statistically significant” under a standard test, even when operationally meaningless. A  $p$ -value here answers the wrong question.

<sup>1</sup>Code: <https://github.com/Custodian-Labs/guardian-layer-phi-benchmark>

3. **Aggregate error hides mechanism.** A single “−1.2 points” says nothing about *why* the loss happens—boundary jitter, detector weakness, or a defect in surrogate generation. Only the last is fixable, and only an error typology tells them apart.

We address all three. Our contributions:

- A **paired multi-detector protocol** scoring utility on masked spans only, decoupling coverage from utility, across 11 detectors  $\times$  7 benchmarks  $\times$  7 languages (§4–§5).
- An **equivalence-testing analysis** (TOST) replacing the uninformative significance test with a bounded-effect claim: the change in detectability is bounded within  $\pm 2$  points of zero (§6).
- A **surrogate-failure typology** attributing the small residual to surrogate-generation quality rather than detectors getting worse at PHI (§7).
- A set of **comparison experiments**—a redaction upper-bound and an open-source surrogate baseline—that make the result reproducible and isolate what is generic to *any* substitution versus specific to one tool (§5, §6).

The framing is deliberately not “a proprietary tool is good.” It is: *here is how to measure whether a structure-preserving transform preserves utility, rigorously*, with one commercial transform as the case study and open baselines for reproducibility.

## 2 Related Work

### 2.1 De-identification and surrogate substitution

Clinical de-identification is a long-studied sequence-labeling problem [31], anchored by the i2b2/n2c2 and MEDDOCAN shared tasks [23, 29] and by systems ranging from rule- and dictionary-based pipelines [25] through recurrent and transformer taggers [8, 16, 20] to instruction-tuned LLMs used zero-shot [21]. Most of this literature optimizes *detection*, typically against the HIPAA Safe Harbor identifier set [30]. The downstream question—*what to put in the PHI’s place*—is comparatively under-studied. Surrogate (“hiding in plain sight”) replacement was proposed to keep de-identified notes realistic and to resist re-identification [4]; subsequent work showed poor surrogates can even aid re-identification (the “parrot” attack), underscoring that surrogate *quality*

matters [5]. Our work is orthogonal to the detection literature and to any particular surrogate generator: given that a span is masked, we ask whether the *replacement* remains detectable and well-formed.

The commercial transform used as our case study is the Custodian Labs Guardian Layer [6]; we cite it as the source of the transform rather than describing or extending the method (a formal method citation will be added once available).

### 2.2 Utility-preservation evaluation

Whether privacy transformations preserve downstream utility is central to privacy-preserving NLP. Prior evaluations typically run a single downstream model on original vs. transformed data and compare end-task scores. Two weaknesses recur: (a) a single downstream model cannot separate “the transform is fine” from “this model is robust,” and (b) whole-corpus metrics mix masked and unmasked content. We address (a) with an 11-detector panel spanning rule-based, fine-tuned, and LLM detectors, and (b) by restricting utility measurement to masked spans.

### 2.3 The large- $N$ trap and equivalence testing

When samples are large, null-hypothesis significance testing rejects the null for negligible effects; the  $p$ -value measures precision, not importance—a hazard increasingly flagged in NLP evaluation [3, 10]. The standard remedy is *equivalence testing*—the two one-sided tests (TOST) procedure [2, 28]—which specifies an equivalence margin  $\Delta$  and tests whether the effect lies inside  $[-\Delta, +\Delta]$ ; see Lakens [17], Lakens et al. [18] for practical tutorials. TOST is standard in biostatistics but under-used in NLP evaluation, where large paired corpora make the trap acute. We adopt it as the primary inferential tool and report McNemar’s paired test [24] only to demonstrate the trap.

## 3 Problem Formulation

**Structure-preserving de-identification.** A transform  $T$  maps a document  $d$  to  $d'$  by replacing each detected PHI value  $v$  (of type  $\tau$ ) with a surrogate  $s$  of the same type, leaving all other characters unchanged. Gold PHI spans on  $d$  are re-projected onto  $d'$  by character-level alignment, giving paired spans  $(v, s)$ .

**Coverage vs. utility.** Two quantities must not be conflated. *Coverage* is the fraction of true PHI that  $T$  detects and replaces (a property of  $T$ ’s detector). *Utility* is, given that a span was replaced, whether a downstream

detector still finds the surrogate  $s$  as well as it found  $v$  (a property of  $T$ ’s generator and of the surrogate’s realism). We report coverage separately and measure utility *only on the masked-span population*  $\{(v, s)\}$ . This prevents a low-coverage transform from looking good—or a high-coverage one from looking bad—on a metric that is really about substitution quality.

## 4 Evaluation Protocol

Figure 1 summarizes the protocol. **Paired multi-detector design.** Each document is scored in two conditions—original  $d$  and transformed  $d'$ —by the *identical* detector suite  $D$  ( $|D| = 11$ ). Because the only change between conditions is the substitution, any per-span change in detection is attributable to the substitution, not to the detector or the document.

**Metrics.** We report span-level P/R/F1 under three matching modes—*exact* (start+end+type), *type* (type+boundary), *overlap* (any character overlap+type)—and *leakage* =  $1 - \text{recall}$ , the HIPAA-critical quantity. The headline utility metric is *recall retention on masked spans* =  $\text{recall}(d')/\text{recall}(d)$  restricted to  $\{(v, s)\}$ , under overlap matching (so pure boundary jitter from length changes, “Anna S.”→“Maria S.”, is not charged as a miss).

**Equivalence testing.** For pooled masked-span recall we run TOST with margin  $\Delta = 2$  points: we reject non-equivalence iff the 90% CI of the recall difference lies entirely within  $[-2, +2]$ . We also report McNemar’s test to illustrate the large- $N$  trap. We sweep  $\Delta \in \{1, 2, 3\}$ .

**Error attribution.** For the lost population (found on  $d$ , missed on  $d'$ ) we hand-code each span into a small failure typology (§7) and check length-preservation to separate boundary artifacts from genuine misses.

## 5 Experimental Design

**Detector panel  $D$ .** Eleven detectors, three families: *rule/statistical*—Microsoft Presidio [25] (built on spaCy [15]), OBI deid\_roberta [16], a RoBERTa tagger [9, 19]; *open LLMs (local)*—Gemma 4 31B / E4B [13], Qwen 3.5- $\{4B, 9B, 35B-A3B\}$  [27], Llama 3.1-8B / 3.3-70B [14], DeepSeek V2-Lite [7]; *frontier API*—OpenAI GPT-5 [26]. All are Transformer-based [32]. The panel is deliberately heterogeneous: if the equivalence result held only for one architecture it would be a model artifact, not a property of the transform. (Model-version citations point to the closest published technical report for each family. One

further model, Moonshot Kimi-VL-A3B, was attempted but excluded because it required non-standard model-loading code.) LLM detectors emit free-text or JSON identifier lists; we recover character spans by exact-then-fuzzy string search over the source document, so all detectors are scored on the same span basis.

**Benchmarks** (7; 250 docs each, 1,750 total). ASQ-PHI (English clinical queries), MEDDOCAN (Spanish clinical) [23], MultiCoNER v2 (multilingual NER) [12, 22], and PII-Masking-300k [1] in English, Dutch, French, German. Together: 7 languages, clinical and general-PII text, free-form and structured (JSON) formats. The transform under test is Custodian Guardian Layer `transform` mode (top-1 surrogate, `pii_entities=ALL`).

**Comparison conditions.** The core study answers “does *this* transform preserve detectability.” Three contrasts isolate *why* and *how generally*:

- **C1—Redaction upper-bound.** Replace each masked value with \*\*\*\*\* (no surrogate signal) and re-detect; the transform—redact gap quantifies what structure preservation buys.
- **C2—Open-source surrogate baseline.** Replace the commercial generator with an open one (Faker [11]) on the *same* masked spans; tests whether the result is generic to substitution or tool-specific, and makes the pipeline reproducible.
- **C3—Per-benchmark equivalence.** Run TOST within each benchmark, sweeping  $\Delta$ , to check the pooled claim is not an averaging artifact.

The core paired study, the equivalence analysis (pooled and per-benchmark), the redaction floor, the open-surrogate baseline, and the error typology are complete; C1/C2 are currently demonstrated with the CPU detector (Presidio), with a full-panel extension noted as the next strengthening step.

## 6 Results

**Detectability is statistically equivalent.** Restricting to the 57,112 masked spans and pooling all 11 detectors (via the released `analyze_equivalence.py`), recall moves 76.1% → 74.9% (−1.2 pts; 95% CI  $[-1.5, -1.0]$ ). The TOST equivalence test ( $\Delta = 2$ ) rejects non-equivalence at  $p \approx 3 \times 10^{-9}$ : the change is statistically bounded within  $\pm 2$  points of zero. Ranking is preserved; Llama 3.3-70B is essentially unchanged

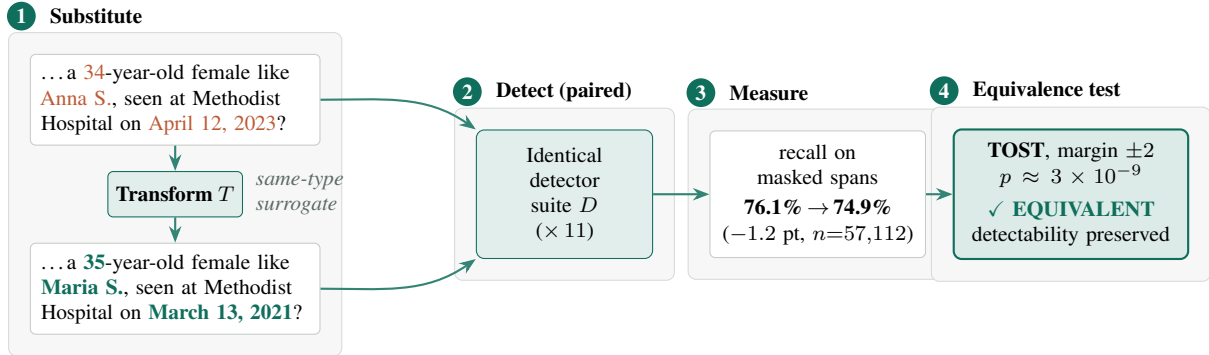


Figure 1: **The paired evaluation protocol, with a worked example.** (1) The transform replaces each PHI value with a same-type surrogate (original → surrogate), leaving everything else byte-identical. (2) The *identical* 11-detector suite scores *both* the original  $d$  and the transformed  $d'$ , so any per-span change is attributable to the substitution alone. (3) Utility is measured *only on the spans the transform masks* (overlap match), decoupling it from coverage. (4) The paired recall difference is assessed with an equivalence test (TOST, margin  $\pm 2$  pts) rather than a significance test—which the large sample would make trivially “significant” (§4). Result: the change is statistically equivalent to zero.

( $\Delta F1 + 0.003$ ). The same contingency is “significant” under McNemar’s test—a direct demonstration of the large- $N$  trap.

**Whole-document view (conservative).** Before restricting to masked spans, Table 1 reports span-level F1 and leakage on *all* gold spans, per detector. Whole-document  $\Delta F1$  ranges from  $+0.003$  (Llama 3.3-70B) to  $-0.047$  (Qwen 3.5-35B-A3B), and leakage rises by only  $+0.3$  to  $+4.1$  points. This view is conservative by design—it mixes masked spans with exact-boundary penalties on length-changed surrogates and with spans the transform never touched—yet it already bounds the worst case (no detector’s mean F1 moves by more than 4.7 points) and preserves ranking across a  $100\times$  span of detector quality (F1  $0.76 \rightarrow 0.04$ ). The masked-span analysis below isolates the substitution effect from this dilution.

**Recall retention on masked spans (overlap).** Restricting to masked spans removes that dilution. When the transform masks a span, detectors still find the surrogate 93–100% of the time (Table 2); the  $\sim 3$ -point exact-boundary drop is a length-jitter artifact that vanishes under overlap matching. (The two floor detectors, Llama 3.1-8B and OBI `deid_roberta`, are omitted from Table 2: their original recall is so low that the retention ratio is dominated by noise.)

**Per-benchmark equivalence (C3).** Table 3 (57,112 masked spans, via `scripts/analyze_equivalence.py`) shows

| Detector                      | F1   |        | $\Delta$ | Leak |        |
|-------------------------------|------|--------|----------|------|--------|
|                               | orig | transf |          | orig | transf |
| Gemma 4 31B                   | .755 | .710   | −.045    | .255 | .285   |
| Gemma 4 E4B                   | .737 | .698   | −.038    | .276 | .311   |
| Llama 3.3-70B                 | .725 | .728   | +0.003   | .245 | .262   |
| Qwen 3.5-35B-A3B              | .715 | .668   | −.047    | .254 | .295   |
| OpenAI GPT-5                  | .705 | .674   | −.032    | .308 | .333   |
| Qwen 3.5-9B                   | .655 | .621   | −.034    | .391 | .422   |
| Qwen 3.5-4B                   | .567 | .533   | −.034    | .483 | .515   |
| Presidio                      | .416 | .398   | −.018    | .553 | .570   |
| DeepSeek V2-Lite              | .409 | .384   | −.025    | .672 | .696   |
| Llama 3.1-8B                  | .391 | .376   | −.015    | .633 | .650   |
| OBI <code>deid_roberta</code> | .041 | .040   | −.002    | .941 | .944   |

Table 1: Whole-document view (all 11 detectors), mean over 7 benchmarks. Span-level F1 (type match) and leakage (1–recall). Conservative: it includes untouched spans and boundary penalties. Ranking is preserved and  $\Delta F1$  is bounded.

the pooled equivalence is *not uniform*: four benchmarks are equivalent within  $\pm 2$  points; MEDDOCAN and PII-nl require  $\pm 3$  (both dense, identifier-heavy, non-English—where surrogate generation is hardest, §7); MultiCoNER has too few masked spans (220) to test. The honest claim: *detectability is equivalent within  $\pm 2$  points on average and on clean text, and within  $\pm 3$  on the hardest identifier-dense text*—the residual is concentrated and attributable, not diffuse degradation

| Detector         | Exact | Overlap      |
|------------------|-------|--------------|
| Gemma 4 31B      | 93.1  | <b>99.8</b>  |
| Qwen 3.5-9B      | 92.0  | <b>100.0</b> |
| Qwen 3.5-35B-A3B | 90.0  | <b>98.4</b>  |
| Llama 3.3-70B    | 91.4  | <b>98.3</b>  |
| GPT-5            | 91.9  | <b>98.3</b>  |
| Gemma 4 E4B      | 88.9  | <b>98.2</b>  |
| Qwen 3.5-4B      | 84.7  | <b>98.0</b>  |
| Presidio         | 91.3  | <b>97.4</b>  |
| DeepSeek V2-Lite | 87.7  | <b>92.8</b>  |

Table 2: Recall retention on masked spans (%), transformed÷original.

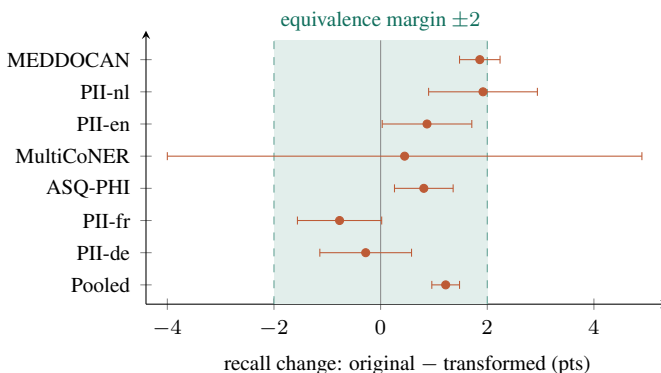


Figure 2: Per-benchmark equivalence (C3). Masked-span recall change (original – transformed) with 95% CIs; the shaded band is the  $\pm 2$ -pt equivalence margin. The pooled estimate and five of seven benchmarks sit fully inside; only MEDDOCAN and PII-nl reach past +2 (hence a  $\pm 3$  margin). MultiCoNER’s wide interval reflects its 220 masked spans.

(Figure 2).

**Redaction floor (C1).** Replacing masked values with \*\*\*\*\* and re-detecting with Presidio (Table 4): transform recall tracks the *original* within a few points, while redaction collapses to  $\approx 0$ –4% (the residual is spurious overlap on the \* run). Structure-preserving substitution preserves essentially the entire detectability that redaction destroys.

**Open-surrogate baseline (C2).** Replacing the commercial generator with Faker on the same masked spans (Table 5) yields two findings. *Generality*: an open substitutor preserves masked-span recall at least as well as the original—detectability preservation appears to be a property of well-formed same-type substitution, not of one vendor, and reproduces without proprietary access.

| Benchmark     | Masked        | O→T              | McN. $\chi^2$ | Margin             |
|---------------|---------------|------------------|---------------|--------------------|
| ASQ-PHI       | 5,427         | 91.9→91.1        | 8.0           | $\pm 2 \checkmark$ |
| MEDDOCAN      | 31,978        | 73.4→71.5        | 92.4          | $\pm 3 \checkmark$ |
| MultiCoNER    | 220           | 71.8→71.4        | 0.0           | underpow.          |
| PII en        | 5,181         | 76.0→75.1        | 4.0           | $\pm 2 \checkmark$ |
| PII nl        | 3,757         | 78.2→76.3        | 13.1          | $\pm 3 \checkmark$ |
| PII fr        | 5,984         | 75.2→75.9        | 3.5           | $\pm 2 \checkmark$ |
| PII de        | 4,565         | 76.3→76.5        | 0.4           | $\pm 2 \checkmark$ |
| <b>Pooled</b> | <b>57,112</b> | <b>76.1→74.9</b> | <b>82.1</b>   | $\pm 2 \checkmark$ |

Table 3: Per-benchmark equivalence. O→T = masked-span recall, original→transform (%). Pooled TOST  $\Delta=2$ :  $p=3\times 10^{-9}$ .

| Benchmark | Masked | Orig. | Transf. | Redact     |
|-----------|--------|-------|---------|------------|
| ASQ-PHI   | 531    | 94.5  | 95.7    | <b>0.0</b> |
| MEDDOCAN  | 2,925  | 75.4  | 72.9    | <b>0.0</b> |
| PII en    | 471    | 75.6  | 73.5    | <b>1.9</b> |
| PII nl    | 373    | 83.6  | 77.7    | <b>3.8</b> |
| PII fr    | 544    | 61.4  | 59.9    | <b>2.2</b> |
| PII de    | 415    | 56.1  | 58.3    | <b>0.0</b> |

Table 4: C1 redaction floor: Presidio masked-span recall (%).

*The residual is generator quality*: Faker, emitting clean canonical values, *exceeds* the commercial transform by 18–20 points on the hard non-English benchmarks, precisely where the commercial surrogates truncate or garble (§7).

**Leakage** barely moves (+0.3 to +4.1 pts across detectors; Table 1), so surrogates are not systematically easier to miss than the PHI they replace.

**Where the loss lands (per benchmark).** Table 6 breaks whole-document  $\Delta F1$  out by benchmark and detector. The pattern is sharp: almost all of the loss concentrates on **ASQ-PHI**—short, sparse-PHI adversarial queries where a single substitution dominates the document score ( $-0.09$  to  $-0.18$ )—while the other six benchmarks are essentially flat ( $|\Delta F1| \leq 0.06$ , most  $\leq 0.03$ ). Even the fine-tuned OBI tagger and the rule-based Presidio move by at most a few thousandths on most benchmarks. This is the first sign that the effect tracks *surrogate-generation difficulty* (short adversarial text, dense non-English identifiers) rather than detector family—made precise in the error analysis (§7).

**What the transform does (examples).** Table 7 shows the substitution qualitatively across languages and

| Benchmark | Masked | Orig. | Custodian | Faker       |
|-----------|--------|-------|-----------|-------------|
| ASQ-PHI   | 531    | 94.5  | 95.7      | <b>97.7</b> |
| MEDDOCAN  | 2,925  | 75.4  | 72.9      | <b>79.7</b> |
| PII en    | 471    | 75.6  | 73.5      | <b>89.2</b> |
| PII nl    | 373    | 83.6  | 77.7      | <b>93.0</b> |
| PII fr    | 544    | 61.4  | 59.9      | <b>78.5</b> |
| PII de    | 415    | 56.1  | 58.3      | <b>78.1</b> |

Table 5: C2 open-surrogate baseline: Presidio masked-span recall (%). Detector-independent argument; Presidio-only demonstration.

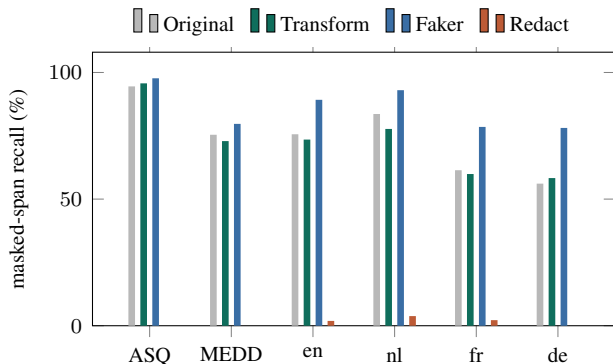


Figure 3: What structure preservation buys (C1/C2; Presidio masked-span recall). Transform tracks the original, the open Faker baseline is at least as high, and redaction collapses to  $\approx 0$ . The gap between the Transform/Faker bars and the Redact bar is the detectability that structure preservation retains.

formats: each PHI value becomes a same-type surrogate while clinical shorthand, foreign-language syntax, and JSON structure are left byte-for-byte intact. On the well-formed English cases every surrogate is still detected across most of the panel.

**Coverage, reported separately.** All results above are measured *on the spans the transform masks*. Coverage—how much true PHI it detects and replaces—is a distinct, detector-side axis. It tracks how closely a benchmark’s annotation matches the transform’s notion of sensitive content: it masks **80.9%** of gold PHI on ASQ-PHI and **48.3%** on MEDDOCAN, and less ( $\approx 26\%$ ) on general-domain NER/PII corpora whose annotated entities (encyclopedic names, generic places) fall outside that scope; a configuration sweep confirmed `domain=General` maximizes coverage. A per-entity-type breakdown and a detection-vs-replacement diagnosis (the coverage gap is mostly a *replacement*-step

issue— $\approx 77\%$  of missed clinical identifiers were flagged by the transform’s own detector but not substituted) are in Appendix A. Decoupling matters because the two failure modes have different owners and fixes: an under-masked span is a *detection/replacement* miss, whereas a masked-but-missed surrogate is a *generation* defect (§7). Conflating them on whole-document F1 (Table 1) would let a high-coverage/low-quality transform and a low-coverage/high-quality one look identical. We therefore report coverage as context and reserve the utility claim for masked spans.

**Matching modes.** Unless noted, retention uses *overlap* matching (detector flags any part of the surrogate span). Exact matching (Table 2, left) additionally requires identical character boundaries and is therefore sensitive to surrogate length changes (“Anna S.”  $\rightarrow$  “Maria S.” shifts the end offset); the  $\sim 3$ -point exact–overlap gap is this boundary jitter, not missed PHI. Type matching (boundary + type) sits between the two and tracks overlap closely.

## 7 Error Analysis

**The lost population.** Pooled across detectors and masked spans: 40,165 (span $\times$ detector) pairs found in both conditions; 3,300 lost. Half (50%) of lost spans are the *same length* as the original—so this is not a boundary effect. By type: LOCATION 27%, NAME 23%, DATE/AGE 22%, ID/contact 19%.

**Three failure modes—all generator-side.** (1) *Malformed/truncated surrogates* (largest cause): *Chicago*  $\rightarrow$  *Illino*, *El Paso*  $\rightarrow$  *El*, *Ciudad de la Habana*  $\rightarrow$  *Cuidad de la Havana*. The fragment no longer matches the lexical pattern detectors learned for real names/places; this is why span-level loss is highest on MEDDOCAN (6.9%; Spanish, dense, identifier-heavy) and lowest on clean English ASQ-PHI (2.5%). (2) *Loss of salience*: a canonical entity replaced by an obscure one (*Cedars-Sinai*  $\rightarrow$  *Vidant*); detectors partly rely on pre-training familiarity, so swapping a famous value for a rare one removes the prior. Inherent to any value substitution; mainly costs weaker detectors. (3) *x-masking of IDs/emails*: *nachorutor@...*  $\rightarrow$  *nxxxxxxxxxx@...*. The *x* run preserves format but breaks the realistic-token pattern. This case is *privacy-positive*—the original value is destroyed—even though it counts against recall.

**Implication.** Detectors are not getting worse at PHI; the small recall gap is driven by surrogate-generation

| Detector         | ASQ-PHI | MEDDOCAN | PII en | PII nl | PII fr | PII de | MultiCoNER |
|------------------|---------|----------|--------|--------|--------|--------|------------|
| Gemma 4 31B      | −.175   | −.017    | −.043  | −.038  | −.014  | −.016  | −.009      |
| Gemma 4 E4B      | −.106   | −.060    | −.032  | −.021  | −.018  | −.025  | −.007      |
| Qwen 3.5-35B-A3B | −.132   | −.034    | −.026  | −.034  | −.040  | −.054  | −.007      |
| OpenAI GPT-5     | −.098   | −.046    | −.049  | −.019  | +0.008 | −.005  | −.014      |
| Qwen 3.5-9B      | −.123   | −.017    | −.025  | −.010  | −.034  | −.024  | −.008      |
| Qwen 3.5-4B      | −.086   | −.055    | −.034  | −.028  | −.015  | −.023  | +0.002     |
| Presidio         | +0.003  | −.037    | −.032  | −.015  | −.020  | −.015  | −.007      |
| Llama 3.1-8B     | −.108   | +0.023   | −.014  | +0.007 | −.009  | +0.003 | −.006      |
| OBI deid_roberta | +0.001  | −.008    | −.004  | −.001  | +0.003 | −.002  | −.001      |

Table 6: Per-benchmark whole-document  $\Delta F1$  (transformed − original; negative = drop). Almost all loss lands on ASQ-PHI; the other six benchmarks are flat. Llama 3.3-70B and DeepSeek V2-Lite omitted for space (both flat; see Table 1 for their pooled  $\Delta F1$ ).

| Case                     | Original                                                                                       | Transformed                                                                                     |
|--------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| ASQ-PHI (clinical query) | ...a 34-year-old female ...like Anna S., ...at Methodist Hospital on April 12, 2023?           | ...a 35-year-old female ...like Maria S., ...at Methodist Hospital on March 13, 2021?           |
| ASQ-PHI (short-hand)     | Rec mgmt of 70yo M w/ CHF, seen by Dr. John L. at Mt. Sinai on Feb 21, 2023. alt tx options... | Rec mgmt of 73yo M w/ CHF, seen by Dr. James L. at Mt. Egypt on Nov 19, 2021. alt tx options... |
| PII (German)             | ...ermächtigen hiermit Monsignore, ...Mit Datum 23/07/2011...                                  | ...ermächtigen hiermit Fulgenzio, ...Mit Datum 24/07/2011...                                    |
| PII (French, JSON)       | { "Date": "20/05/2022", "City": "Saint-Priest", "Username": "phprosdocimo" }                   | { "Date": "21/05/2023", "City": "Saint-Priest", "Username": "phprosdocimo" }                    |

Table 7: Worked transform examples. Only the sensitive value moves; abbreviations (w/ CHF, alt tx), non-English syntax, and JSON keys/quotes/indentation are preserved, so downstream parsers and detectors keep working. In the JSON case the transform changes only the date value and leaves the (also-sensitive-looking) username untouched—a coverage decision, not a utility one (see below).

quality (truncation, garbling, salience, x-masking). C2 confirms this directly: an open generator emitting clean canonical values *exceeds* the commercial transform by 18–20 points on exactly the hard non-English benchmarks where its surrogates truncate—so the residual tracks generator quality, not the act of substitution.

## 8 Discussion

**What the evidence supports.** Structure-preserving substitution does not hide well-formed PHI from downstream detection. Because the effect is equivalence-bounded within  $\pm 2$  points and ranking-preserving across a heterogeneous 11-detector panel, a transform of this quality can be inserted ahead of detection/analytics pipelines built for real clinical text without materially de-

grading downstream detection. **Generality (C2).** Faker preserves masked-span recall at least as well as the original across all six benchmarks, so  $\pm 2$  points appears to be a property of well-formed substitution rather than of one vendor; where the commercial generator trails, a cleaner generator closes the gap, locating the residual squarely in generation quality. **A reusable protocol.** The paired masked-span design and the TOST margin are not specific to one vendor or language—a template for auditing any structure-preserving privacy transform.

## 9 Ethics and Data Statement

All benchmarks are public or synthetic (ASQ-PHI synthetic; MEDDOCAN released for a shared task; PII-Masking-300k synthetic; MultiCoNER v2 public). **No**

**real patient data is used.** We release the 250-document subsets and scoring code with license notes. **Dual-use.** A utility-preserving de-identifier could in principle launder identifiable data into a fluent form; our masked-span/leakage reporting and the privacy-positive framing of  $x$ -masking keep the privacy accounting explicit. **Conflict of interest.** The commercial transform (Custodian Guardian Layer) is developed by Custodian Labs, with which the authors are affiliated; this work was conducted with Custodian Labs’ support. To limit bias we (i) frame the contribution as a reusable protocol, (ii) include open-source baselines so results are reproducible without proprietary access, and (iii) report leakage and coverage alongside utility.

## 10 Limitations

Coverage is reported but not the focus; a transform can preserve utility on what it masks while under-masking (the axes are independent by design). 250 docs/benchmark bounds per-benchmark power (though the pooled masked-span  $N$  is large). C1/C2 are currently demonstrated with a CPU detector (Presidio); a full 11-detector extension is future work. LLM detectors are prompt-sensitive; we fix one prompt per model and release it with the code and data at the project’s reproducibility page. Finally, all seven benchmarks are public or synthetic; validating the protocol on real-EHR corpora (n2c2, MIMIC-IV-Note, CARMEN-I) is future work pending the relevant data-use agreements.

### A Coverage Details

Coverage (§6) is the fraction of a benchmark’s gold PHI whose characters the transform actually changed (diff/alignment). Table 8 breaks it down by entity type on the clinical sets (ASQ-PHI + MEDDOCAN). High-frequency free-text types (names, dates) are masked well; the weak spots are high-sensitivity structured identifiers and geography—exactly the HIPAA Safe Harbor items that most need masking.

**Detection vs. replacement.** On a sample of 8 MEDDOCAN documents (62 missed ID/location spans), 48/62 ( $\approx 77\%$ ) of the *unmasked* identifiers were nonetheless flagged as sensitive by the transform’s own detector—they were detected but not substituted; only 14/62 (23%) were undetected. The coverage gap is therefore mostly a *replacement-step* issue (act on everything the detector surfaces), which is more tractable than raising detection recall. (Small sample; “de-

| Entity type (clinical)                       | Coverage |
|----------------------------------------------|----------|
| NAME / person                                | 75.0%    |
| DATE / age                                   | 73.4%    |
| ID / contact (MRN, patient ID, phone, email) | 50.7%    |
| ORG / facility                               | 46.6%    |
| LOCATION / address                           | 43.7%    |

Table 8: Coverage by entity type on clinical benchmarks. Overall the transform altered 39% of annotated PII (54% on clinical sets); general-domain corpora sit near 26%.

tected” judged by loose token overlap, so 77% is directional.) Concrete unmasked HIPAA identifiers included patient IDs (80926, 7845693), care-contact IDs (4387684), and facilities/geography (Hospital de Cruces, España, postal codes 41005, 28047).

### B Representative Per-Detector Examples

**All-detector overlay on one document (“ranking proof”).** All eleven detectors’ predictions on one transformed document (asq\_00001), overlaid on the text and sorted by this document’s F1: **green** = correctly caught PHI, **yellow** = false positive, **red** = missed PHI (leakage). The four gold surrogate values are John T., the  $x$ -masked location Saint. Vxxxxxxxx, April 25th, 2018, and the ID 987654321. Eight of eleven detectors tag all four cleanly (F1 100); the two floor detectors, Presidio and DeepSeek, miss the  $x$ -masked location surrogate (§7 mode 3)—DeepSeek also drops the numeric ID—and OBI over-fragments boundaries and false-fires on the age 60. The surrogate values shown are the transform’s output, not the original PHI.

|                                                                                                                                                                     |                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Gold (reference)</b>                                                                                                                                             | 4 PHI spans                      |
| Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxxx on April 25th, 2018 (ID: 987654321)? |                                  |
| <b>OpenAI GPT-5</b>                                                                                                                                                 | tp 4 · fp 0 · missed 0 · F1 100% |
| Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxxx on April 25th, 2018 (ID: 987654321)? |                                  |
| <b>Llama 3.1-8B</b>                                                                                                                                                 | tp 4 · fp 0 · missed 0 · F1 100% |
| Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxxx on April 25th, 2018 (ID: 987654321)? |                                  |

|                                                                                                                                                                    |                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Qwen 3.5-4B</b>                                                                                                                                                 | tp 4 · fp 0 · missed 0 · F1 100% |
| Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)? |                                  |
| <b>Gemma 4 31B</b>                                                                                                                                                 | tp 4 · fp 0 · missed 0 · F1 100% |
| Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)? |                                  |
| <b>Llama 3.3-70B</b>                                                                                                                                               | tp 4 · fp 0 · missed 0 · F1 100% |
| Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)? |                                  |
| <b>Qwen 3.5-35B-A3B</b>                                                                                                                                            | tp 4 · fp 0 · missed 0 · F1 100% |
| Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)? |                                  |
| <b>Gemma 4 E4B</b>                                                                                                                                                 | tp 4 · fp 0 · missed 0 · F1 100% |
| Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)? |                                  |
| <b>Qwen 3.5-9B</b>                                                                                                                                                 | tp 4 · fp 0 · missed 0 · F1 100% |
| Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)? |                                  |
| <b>OBI deid_roberta</b>                                                                                                                                            | tp 4 · fp 1 · missed 0 · F1 89%  |
| Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)? |                                  |
| <b>Microsoft Presidio</b>                                                                                                                                          | tp 3 · fp 1 · missed 1 · F1 75%  |
| Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)? |                                  |
| <b>DeepSeek V2-Lite</b>                                                                                                                                            | tp 2 · fp 0 · missed 2 · F1 67%  |
| Evaluation of long-term outcomes for bypass surgery in patients over 60, referencing Mr. John T., operated at Saint. Vxxxxxxx on April 25th, 2018 (ID: 987654321)? |                                  |

### A harder case: a dense Spanish clinical header.

Figure-style overlay on a 260-character window of a MEDDOCAN document (S0210-...008-1), where the same eleven detectors diverge far more—F1 from 96 down to 0. On dense, identifier-heavy non-English text the *per-document* ranking reshuffles (here DeepSeek and Qwen-4B lead, while Llama 3.1-8B finds nothing in this window), even though the *aggregate* ranking over 250 documents is stable (Table 1). The garbled surrogate Ciudad Real (from Ciudad Real) is still caught by most detectors; the misses cluster on the professional-licence ID 03 14 16485, the sex fields, and the re-

peated age—the identifier types with the weakest coverage and the hardest surrogates (Appendix A).

|                                                                                                                                                                                                                                                                      |                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Gold (reference)</b>                                                                                                                                                                                                                                              | 12 PHI spans                     |
| Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q |                                  |
| <b>DeepSeek V2-Lite</b>                                                                                                                                                                                                                                              | tp 11 · fp 0 · missed 1 · F1 96% |
| Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q |                                  |
| <b>Qwen 3.5-4B</b>                                                                                                                                                                                                                                                   | tp 11 · fp 0 · missed 1 · F1 96% |
| Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q |                                  |
| <b>OBI deid_roberta</b>                                                                                                                                                                                                                                              | tp 10 · fp 0 · missed 2 · F1 91% |
| Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q |                                  |
| <b>Qwen 3.5-9B</b>                                                                                                                                                                                                                                                   | tp 10 · fp 0 · missed 2 · F1 91% |
| Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q |                                  |
| <b>Gemma 4 31B</b>                                                                                                                                                                                                                                                   | tp 9 · fp 0 · missed 3 · F1 86%  |
| Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q |                                  |
| <b>Qwen 3.5-35B-A3B</b>                                                                                                                                                                                                                                              | tp 9 · fp 1 · missed 3 · F1 82%  |
| Ciudad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q |                                  |

|                                                                                                                                                                                                                                                                      |                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>OpenAI GPT-5</b>                                                                                                                                                                                                                                                  | tp 8 · fp 0 · missed 4 · F1 80% |
| Cuidad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q |                                 |
| <b>Llama 3.3-70B</b>                                                                                                                                                                                                                                                 | tp 8 · fp 0 · missed 4 · F1 80% |
| Cuidad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q |                                 |
| <b>Gemma 4 E4B</b>                                                                                                                                                                                                                                                   | tp 8 · fp 0 · missed 4 · F1 80% |
| Cuidad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q |                                 |
| <b>Microsoft Presidio</b>                                                                                                                                                                                                                                            | tp 6 · fp 1 · missed 6 · F1 63% |
| Cuidad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q |                                 |
| <b>Llama 3.1-8B</b>                                                                                                                                                                                                                                                  | tp 0 · fp 0 · missed 12 · F1 0% |
| Cuidad Real. CP: 13002. Datos asistenciales. Referencia de nacimiento: 24/10/1963. País: España. Mejores: 53 años Sexo: H. Referencia de Ingreso: 14/12/2017. Episodio: 746589123. Médico: Luis Ruiz Camuñas N°Col: 03 14 16485. Historia Actual: Varón de 53 años q |                                 |

**Agreement on well-formed surrogates.** On clean English (ASQ-PHI), a plausible same-type surrogate is caught by nearly the whole panel (Table 9): the substitution is invisible to detection. Divergence is confined to the two floor detectors (Llama 3.1-8B, DeepSeek V2-Lite).

| Document  | Surrogate span (type)    | Caught |
|-----------|--------------------------|--------|
| asq_00000 | Maria S. (NAME)          | 10/11  |
| asq_00000 | Methodist Hospital (LOC) | 9/11   |
| asq_00000 | March 13, 2021 (DATE)    | 9/11   |
| asq_00003 | James L. (NAME)          | 11/11  |
| asq_00003 | Mt. Egypt (LOC)          | 11/11  |
| asq_00003 | Nov 19, 2021 (DATE)      | 11/11  |

Table 9: Number of detectors (of 11) that find each well-formed surrogate. Same-type swaps stay broadly detectable.

**Representative losses, by failure type.** Table 10 shows genuine “lost” cases (MEDDOCAN): the original value was found by most detectors, but a defective surrogate is found by few. Each maps to one of the three failure types in §7, and the drop is shared across the panel—i.e. it is a property of the surrogate, not of any one detector.

| Original → surrogate                | O → T  | Type            |
|-------------------------------------|--------|-----------------|
| Cuba → Havana                       | 9 → 2  | salience        |
| San Fernando → Francsico Luis       | 8 → 3  | garbled         |
| Cádiz → Cadiz                       | 8 → 4  | accent stripped |
| Cecilio                             | 8 → 4  | truncation      |
| Pujazón → Pujazón                   |        |                 |
| ignaciotorne@... → ignaciotorne@... | 9 → 2  | x-mask          |
| ixxxxxxxxxxxxx@...                  |        |                 |
| 23/07/1948 → 2xxxxxxxx              | 11 → 4 | x-mask          |
| 40 → 35 (age)                       | 9 → 2  | bare number     |

Table 10: Representative lost spans on MEDDOCAN. O → T = detectors finding the original vs. the surrogate (panel of 11–12; original includes one extra decoding variant). Losses are shared across detectors and align with the §7 typology.

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